Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: *LTZGLUE* | ***Context:*** *Luxembourg Public Administration — Civil Servant NLU Deployment*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is text-classification labels evaluated by macro-F1 with results averaged over three runs and standard deviations reported [Q93, Q96]. This is a standard, well-documented evaluation methodology and matches the deployment's text-to-label format at surface level. However, the deployment requires confidence scores for routing thresholds and multi-label outputs for messages spanning multiple departments [elicitation Q4]. LTZGLUE reports only macro-F1 with no calibration metrics (e.g., ECE, Brier score), no per-class precision/recall thresholds, and no multi-label evaluation methodology. Invalid LLM outputs are simply discarded [Q90, Q91], which masks the operational reality that malformed outputs in deployment must be handled (escalated). OF is MODERATE priority. The form is text-to-text-label as required, but the metric set and label cardinality do not match the deployment's operational form.

ID	Evidence
Q4	"Small and under-researched languages are particularly difficult to evaluate, as is the case with Luxembourgish (LTZ), the national language of Luxembourg, with around 400k speakers." (p.1)
Q90	"Prompted LLMs do not always produce well-formed outputs and may return an incorrect number of predictions for a given task; such outputs are discarded prior to evaluation." (p.7)
Q91	"All reported scores are computed on the remaining valid predictions per model." (p.7)
Q93	"Table 6 shows F1 scores for all models across all tasks (see Appendix 7.9 for full results)." (p.7)
Q96	"Table 6: Test F1 scores across all ltzGLUE tasks. Encoder results are averaged over three runs with standard deviations as subscripts. Prompted LLMs were evaluated once; we report macro-F1 only." (p.8)

Strengths

- Text-output modality matches deployment [Q93]
- Multi-run evaluation with standard deviations reported for encoder models [Q89, Q96] supports meaningful comparison
- Hyperparameter search via Bayesian optimization with early stopping is methodologically sound [Q136, Q138]
- Class-balanced loss for imbalanced tasks [Q92] is a conscientious design choice

ID	Evidence
Q89	"For encoder-based models, results are reported as averages over multiple runs (see Appendix 7.2 for more details)." (p.7)
Q92	"For the supervised models, since the linguistic acceptability and sentiment analysis datasets are highly imbalanced, when fine-tuning on these tasks we use class-balanced loss based on effective size (Cui et al., 2019) with a beta of 0.99." (p.7)
Q93	"Table 6 shows F1 scores for all models across all tasks (see Appendix 7.9 for full results)." (p.7)
Q96	"Table 6: Test F1 scores across all ltzGLUE tasks. Encoder results are averaged over three runs with standard deviations as subscripts. Prompted LLMs were evaluated once; we report macro-F1 only." (p.8)
Q136	"Though we do not aim to optimise performance in our evaluation, we conduct basic hyperparameter sweeps for each model and task combination in order to provide a fairer comparison across models." (p.13)
Q138	"In order to reduce the computational demand of the sweeps, we use Bayesian search with early stopping after three iterations, and cap each sweep at 30 runs, for 1,440 total runs across all models and tasks (and an additional 96 to finetune the two additional seeds)." (p.13)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Text output matches deployment, but the deployment additionally requires confidence scores and multi-label outputs [elicitation Q4], which the benchmark's macro-F1 single-label evaluation does not assess.

OF-2: Not applicable — deployment is text-only [regional context infrastructure_notes.deployment_modality], and benchmark is text-only [Q93].

OF-3: Civil servants are professionally literate [regional context target_population.civil_servant_profile]; literacy is not a constraint. Citizens have 99.0% internet penetration [WEB-7]. No accessibility-driven output modality concerns.

OF-4: External-validity gap: macro-F1 alone cannot validate the calibration and multi-label behaviors required for routing escalation logic; invalid outputs being discarded [Q90, Q91] hides a deployment-relevant failure mode.

ID	Evidence
Q4	“Small and under-researched languages are particularly difficult to evaluate, as is the case with Luxembourgish (LTZ), the national language of Luxembourg, with around 400k speakers.” (p.1)
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WEB-7	99.0% internet penetration — confirms text-output channel is universally accessible (https://datareportal.com/reports/digital-2024-luxembourg)

Information Gaps

- No calibration metrics (ECE, reliability diagrams) reported
- No multi-label evaluation methodology defined
- No analysis of malformed-output rate as an operational signal

Requires Expert Verification

- Specification of operational confidence thresholds and escalation-rate targets by deployment owners

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Remediation

Gap 1: Macro-F1 only; no calibration metrics; invalid outputs are silently discarded; no multi-label evaluation

Recommendation: Add calibration metrics (ECE, reliability diagrams), multi-label metrics (micro/macro-F1, hamming loss, subset accuracy), and treat malformed-output rate as an explicit deployment KPI rather than discarding such cases.